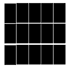
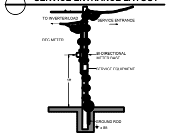


GENERAL NOTES AND SPECIFICATIONS

1. ALL ELECTRICAL WORKS SHALL BE DONE IN ACCORDANCE WITH THE LATEST EDITION OF PHILIPPINE ELECTRICAL CODE.
2. MINIMUM SIZE OF WIRE AND CONDUIT FOR PHOTOVOLTAIC MODULE SHALL BE 4mm^2 DC CABLE AND 15mm Ø RESPECTIVELY UNLESS OTHERWISE SPECIFIED ON PLAN.
3. ALL MATERIALS AND EQUIPMENT TO BE USED SHALL BE BRAND NEW AND APPROVED TYPE FOR ALL LOCATION AND PURPOSES.
4. ALL METAL FRAMES SHALL BE PROPERLY AND ADEQUATELY GROUNDED.
5. ALL WEATHER-EXPOSED INSTALLATIONS SHALL USE WEATHERPROOF TYPE MATERIALS, ESPECIALLY WEATHERPROOF CONDUITS, JUNCTION BOXES AND SAFETY DEVICES.
6. ALL SPECIAL EQUIPMENT (INVERTERS, PV MODULE AND BATTERY) SHALL BE INSTALLED IN ACCORDANCE WITH STANDARDS SET BY THE CODE AND MANUFACTURER'S SPECIFICATIONS.
7. BATTERIES AND INVERTERS SHALL BE INSTALLED IN A WELL SECURED AND VENTILATED LOCATION.
8. ALL INSTALLATION SHALL BE DONE IN GOOD AND NEAT WORKMANSHIP.
9. ALL ELECTRICAL WORKS SHALL BE DONE UNDER THE DIRECT SUPERVISION OF REGISTERED ELECTRICAL ENGINEER/REGISTERED MASTER ELECTRICIAN OR PROFESSIONAL SOLAR SYSTEM INSTALLER.

SERVICE ENTRANCE LAYOUT



SHORT CIRCUIT ANALYSIS

TRANSFORMER RATING = 75KVA, SINGLE PHASE, %Z = 5, Vsec = 230V

FAULT X1

$$IFLA = \frac{75KVA \times 1000}{230V}$$

IFLA = 326.09A

$$MULTIPLIER = \frac{100}{TRANSFORMER\ IMPEDANCE \times 0.9}$$

$$MULTIPLIER = \frac{100}{5 \times 0.9} = 22.22$$

SHORT CIRCUIT CURRENT @ SECONDARY OF TRANSFORMER

I(S.C.)(L-L)(X1)= 326.09A x 22.22 = 7245.72A

SHORT CIRCUIT CURRENT @ SECONDARY OF TRANSFORMER

FAULT X2

$$\frac{2 \times L \times I(S.C.)}{C \times n \times VSEC}$$

WHERE:
L = LENGTH (FEET) OF CONDUCTOR TO THE FAULT
C = CONSTANT FROM TABLE 4 OF "C" VALUES FOR CONDUCTORS
n = NUMBER OF CONDUCTORS PER PHASE
I = AVAILABLE SHORT CIRCUIT CURRENT IN AMPERES AT THE BEGINNING OF CIRCUIT
V = VOLTAGE OF CIRCUIT

$$\frac{2 \times (50 \times 3.28) \times 7245.72A}{1557 \times 1 \times 230V} = 6.64$$

$$M = \frac{1}{1+f} = \frac{1}{1+6.64} = 0.13$$

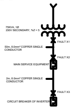
I(S.C.)(L-L)(X2)= M x I(S.C.)(L-L)(X1)
= 0.13 x 7245.72A
= 948.39A

FAULT X3

$$\frac{2 \times (2 \times 3.28) \times 948.39A}{1557 \times 1 \times 230V} = 0.03$$

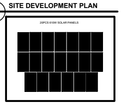
$$M = \frac{1}{1+f} = \frac{1}{1+0.03} = 0.97$$

I(S.C.)(L-L)(X3)= M x I(S.C.)(L-L)(X2)
= 0.97 x 948.39A
= 920.76A

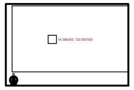

75KVA, 1Ø
230V SECONDARY, %Z = 5
FAULT X1
50m, 8.0mm² COPPER SINGLE CONDUCTOR
FAULT X2
MAIN SERVICE EQUIPMENT
2m, 8.0mm² COPPER SINGLE CONDUCTOR
FAULT X3
CIRCUIT BREAKER OF INVERTER

THEREFORE, SELECT KAIC RATING OF CIRCUIT BREAKER NOT LESS THAN 10KAIC

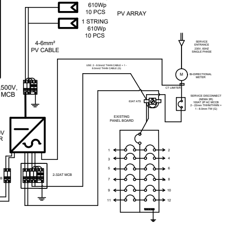
SITE DEVELOPMENT PLAN



VICINITY MAP



SINGLE LINE DIAGRAM



VOLTAGE DROP CALCULATION

MAIN FEEDER WIRE: 100AT MCCB

1.) 2-22.00mm^2 THHN
R = 0.122 ohms
L = 85ft
I = 125.02 A

$$VD = \frac{2 \times L \times R \times I}{1000}$$

$$VD = \frac{2 \times 85\ ft \times 0.122\ ohms/ft \times 125.02\ A}{1000}$$

VOLTAGE DROP = 2.59VOLTS

$$9\ VOLTAGE\ DROP = \frac{3.43VOLTS}{230\ volts} \times 1000$$

%VOLTAGE DROP = 1.20%

TRANSFORMER SIZING : 150AT CB

KVA = I x 230
= 125.02 x 230
= 28,754.60 W

HYBRID INVERTER SPECIFICATION

MODEL	-	DEYE 8KW SG05LP1-EU-SM2
BATTERY TYPE	-	LEAD ACID OR LITHIUM IRON
BATTERY VOLTAGE RANGE (V)	-	40 - 60
MAX CHARGING CURRENT (A)	-	190A
MAX DISCHARGING CURRENT (A)	-	190A
MAX. DC INPUT POWER	-	16000W
MAX. DC INPUT VOLTAGE	-	500V
MPPT VOLTAGE RANGE	-	150-425 V
MAX. DC INPUT CURRENT	-	26A+26A
MAX. SHORT CIRCUIT	-	34A+34A
RATED AC GRID VOLTAGE	-	220/230 0.85Un-1.1Un
RATED AC GRID FREQUENCY	-	50/45-55, 60/55-65 HZ
RATED AC OUTPUT POWER	-	8000W
MAX. APPARENT POWER	-	88000VA
MAX. AC OUTPUT CURRENT	-	38.3A
POWER FACTOR	-	0.8 Leading - 0.8 Lagging

UNITS/ABBREVIATIONS

A - AMPERE
AT - AMPERE TRIP
AF - AMPERE FRAME
V - VOLTAGE
VA - VOLT AMPERE
KW - KILOWATT
MCB - MINIATURE CIRCUIT BREAKER
AC - ALTERNATING CURRENT
DC - DIRECT CURRENT
PV - PHOTOVOLTAIC
2P - TWO POLE/DOUBLE POLE
SPD - SURGE PROTECTION DEVICE
KAIC - KILO AMPERE INTERRUPTING CAPACITY

LEGENDS AND SYMBOLS

HYBRID INVERTER
PV MODULE
GROUND
BI DIRECTIONAL METER
DC SPD/AC SPD
AUTOMATIC TRANSFER SWITCH
DC BREAKER
DC/AC BREAKER

PANEL STRING CALCULATION

1ST STRING SERIES CONNECTION (10S)

610W SOLAHESTIA SOLAR PANELS
VOC = 48.0V
VMP = 40.8V
IMP = 14.96A

TOTAL PV VOLTAGE OUTPUT(VOC) = 48.0V x 10 = 480V
TOTAL PV VOLTAGE OUTPUT (VMP) = 40.8V x 10 = 408V
TOTAL PV OUTPUT CURRENT (IMP) = 14.96A

2ND STRING SERIES CONNECTION (10S)

610W SOLAHESTIA SOLAR PANELS
VOC = 48.0V
VMP = 40.8V
IMP = 14.96A

TOTAL PV VOLTAGE OUTPUT(VOC) = 48.0V x 10 = 480V
TOTAL PV VOLTAGE OUTPUT (VMP) = 40.8V x 10 = 408V
TOTAL PV OUTPUT CURRENT (IMP) = 14.96A

PROFESSIONAL ELECTRICAL ENGINEER
PRC NO :
PTR NO :
PLACE :
VALIDITY :
DATE :
TIN :

SHEET CONTENT
PV MODULE LAYOUT
VICINITY MAP
PV SYSTEM SINGLE LINE DIAGRAM
PV SYSTEM DESIGN ANALYSIS
VOLTAGE DROP CALCULATION
DESIGN BY :
CAD BY :
DATE :

PROJECT TITLE
12.20KWP HYBRID SOLAR PV
ROOFTOP SYSTEM
WITH NET METERING

PROJECT LOCATION
0017 LT BLOCK 1 ST
PONTICELLI MOLINO III
BACOOR CAVITE

OWNER
TYRONE P. MATIAS

SHEET NO.
E1
1

SCHEDULE OF LOADS

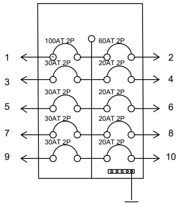
CKT NO.	DESCRIPTION	VA LOAD PER CKT	CIRCUIT BREAKER AF	AT	P	WIRES mm^2 THHN, mm^2 TW (G)	CONDUIT mmØ PVC
1	MAIN	2500	100	100	2	2-8.0mm^2, 1-2.0mm^2	20
2	RANGE HOOD	1500	20	20	2	2-5.5mm^2, 1-2.0mm^2	20
3	CONVENIENCE OUTLET	1300	16	16	2	2-5.5.0mm^2, 1-2.0mm^2	20
4	LIGHTING OUTLET	600	16	16	2	2-5.5.0mm^2, 1-2.0mm^2	20
5	LIGHTING OUTLET	600	16	16	2	2-5.5mm, 1-2.0mm	20
6	ACU	1840	32	32	2	2-5.5mm^2, 1-2.0mm^2	20
7	ACU	1840	32	32	2	2-5.5mm^2, 1-2.0mm^2	20
8	ACU	1840	32	32	2	2-5.5mm^2, 1-2.0mm^2	20
9	REFRIGERATOR	1400	32	32	2	2-5.5mm^2, 1-2.0mm^2	20
10	ACU	1840	32	32	2	2-5.5mm^2, 1-2.0mm^2	20
	TOTAL VA LOAD	15,260					

DEMAND LOAD IN AMPERES :

$$I_L @\ 70\%\ D.F. = \frac{15,260 \times 70\%}{230} + (0.25 \times 10) = 48.94\ AMPERES$$

MAIN FEEDER : 2 - 22mm^2 THHN/THWN + 1 - 8.0mm^2 TW (G) IN 1" RSC

LOAD SCHEDULE



RISER DIAGRAM

PV DATASHEET

PROFESSIONAL ELECTRICAL ENGINEER
PRC NO :
PTR NO :
PLACE :
VALIDITY :
DATE :
TIN :

SHEET CONTENT
RISER DIAGRAM
LOAD SCHEDULE
DESIGN BY :
CAD BY :
DATE :

PROJECT TITLE
12.20KWP HYBRID SOLAR PV
ROOFTOP SYSTEM
WITH NET METERING

PROJECT LOCATION
0017 LT BLOCK 1 ST
PONTICELLI MOLINO III
BACOOR CAVITE

OWNER
TYRONE P. MATIAS

SHEET NO.
E1
2